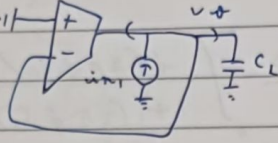


|             |                |
|-------------|----------------|
| Name        | Amogh G. Okade |
| Roll number | EP22B020       |

6.1:

classmate  
Date \_\_\_\_\_  
Page \_\_\_\_\_

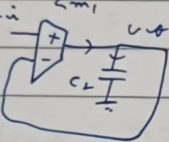
1.4.26 Assignment - 6

4.6.1 a) 

$$i_{n1} = g_{m1} v_{in1} + v_{out} s C_L$$

$$\therefore \frac{i_{n1}}{g_{m1} + s C_L} = v_{out}$$

$$\therefore s v_{out} = \frac{s i_{n1}}{g_{m1}^2 + (2\pi f C_L)^2}$$

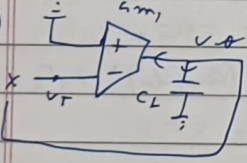
→ 

$$g_{m1} (v_{in1} - v_{out}) = v_{out} s C_L$$

$$\therefore v_{in1} \times g_{m1} = v_{out} (g_{m1} + s C_L)$$

$$\therefore s v_{in1} = \frac{s v_{out} [g_{m1}^2 + (2\pi f C_L)^2]}{g_{m1}^2}$$

→  $\therefore s v_{in1} = \frac{s i_{n1}}{g_{m1}^2} = \frac{4\pi f g_{m1}}{g_{m1}} = \frac{4\pi f}{g_{m1}} = s v_{in1}$

2) 

$$g_{m1} v_T + v_{out} s C_L = 0$$

$$\therefore \angle G(s) = \frac{v_{out}}{v_T} = -\frac{g_{m1}}{s C_L}$$

DC gain = 1  $\left\{ \begin{array}{l} \omega_{u, loop} = \frac{g_{m1}}{C_L} \end{array} \right\}$  not  $\phi_2$  and  $z_1$

3) 
$$s v_{out} = \frac{4\pi f g_{m1}}{g_{m1}^2 + (2\pi f C_L)^2} = \frac{4\pi f}{C_L} \left( \frac{4 g_{m1} C_L}{g_{m1}^2 + (2\pi f C_L)^2} \right)$$

classmate

Date \_\_\_\_\_

Page \_\_\_\_\_

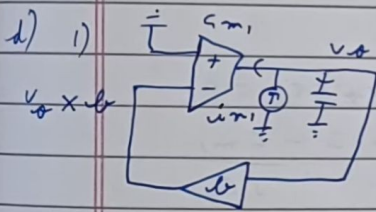
$$\sigma^2 = \int_0^{\infty} S_{v_{n\phi}} df = \frac{\pi J}{C_L} \times \frac{4g_{m1}C_L}{g_{m1}} \int_0^{\infty} \frac{df}{1 + \left(\frac{2\pi f C_L}{g_{m1}}\right)^2}$$

$$= \frac{\pi J}{C_L} \times \frac{4C_L}{g_{m1}} \times \frac{g_{m1}}{2\pi C_L} \int_0^{\infty} \frac{d(2\pi f C_L / g_{m1})}{1 + (2\pi f C_L / g_{m1})^2}$$

$$\sigma^2 = \frac{\pi J}{C_L} \times \frac{2}{\pi} \times \frac{\pi}{2} = \frac{\pi J}{C_L}$$

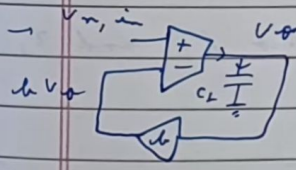
4)  $NBW = \frac{\pi J / C_L}{\frac{4\pi J}{g_{m1}} \times 1^2} = \frac{g_{m1}}{4C_L}$

5)  $S_{v_{n, in}} = \frac{4\pi J}{\omega_{u, loop} \times C_L}$

d) 1) 

$$i_{n1} = v_{\phi} s C_L + g_{m1} v_{\phi} dr$$

$$\therefore S_{v_{n\phi}} = \frac{S i_{n1}}{(g_{m1} dr)^2 + (2\pi f C_L)^2}$$

2) 

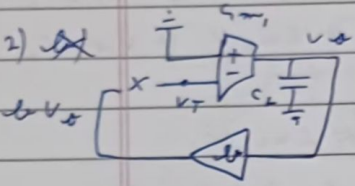
$$g_{m1} (v_{n, in} - dr v_{\phi}) = v_{\phi} s C_L$$

$$\therefore g_{m1} v_{n, in} = v_{\phi} (s C_L + dr g_{m1})$$

$$\therefore S_{v_{n, in}} = \frac{S_{v_{n\phi}} [(g_{m1} dr)^2 + (2\pi f C_L)^2]}{g_{m1}^2}$$

classmate  
Date \_\_\_\_\_  
Page \_\_\_\_\_

$$\therefore S v_{n, in} = \frac{S i_{n1}}{g_{m1}} = \frac{4kT}{g_{m1}} = S v_{n1}$$

2) ~~✗~~ 

$$g_{m1} V_T + v_o S C_L = 0$$

$$\therefore \frac{v_o}{v_T} = A(s) = -\frac{g_{m1}}{S C_L}$$

$$L_A(s) = \ln A(s) = \frac{-\ln g_{m1}}{S C_L}$$

$$DC \text{ gain} = \frac{1}{\ln} \left\{ \omega_{u, loop} = \frac{\ln g_{m1}}{C_L} \right\} \text{ No of } z, 2, 1$$

3) ~~✗~~ 
$$S v_{n+} = \frac{kT}{C_L} \times \frac{4 g_{m1} C_L}{(g_{m1} \ln)^2 + 2\pi f C_L} \times \frac{\ln}{\ln}$$

$$\therefore \sigma^2 = \int_0^\infty S v_{n+} df = \frac{kT}{C_L} \times \frac{1}{\ln}$$

4) 
$$NBW = \frac{kT}{C_L \ln} = \frac{g_{m1} \times \ln}{4 C_L}$$

$$\frac{4 kT \times (1)^2}{g_{m1} (\ln)}$$

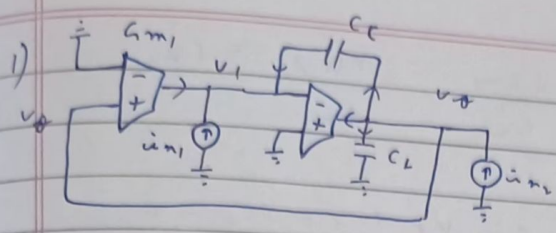
5) ~~✗~~ 
$$S v_{n, in} = \frac{4 kT \times \ln}{C_L \times \omega_{u, loop}} = \frac{4 kT}{A_0 C_L \times \omega_{u, loop}}$$

classmate

Date \_\_\_\_\_

Page \_\_\_\_\_

2) 1)



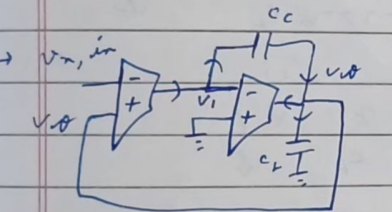
$G_{m1}$   
 $v_i$   
 $v_o$   
 $i_{n1}$   
 $C_c$   
 $C_L$   
 $i_{n2}$

$G_{m1} v_o + i_{n1} + (v_o - v_i) s C_c = 0$   
 $i_{n2} = (v_o - v_i) s C_c + G_{m2} v_i + v_o s C_L$

Solving,  $v_o = \frac{i_{n1} (s C_c - G_{m2}) + i_{n2} s C_c}{s^2 C_c C_L + s C_c (G_{m2} - G_{m1}) + G_{m1} G_{m2}}$

$s v_{n,o} = \frac{s i_{n1} (\omega^2 C_c^2 + G_{m2}^2) + s i_{n2} \omega^2 C_c^2}{\omega^2 C_c^2 (G_{m2} - G_{m1})^2 + (G_{m1} G_{m2} - \omega^2 C_c C_L)^2}$   
 $s v_{n,o} = \frac{s i_{n1} [G_{m2}^2 + \omega^2 C_c^2 (1 + G_{m2}/G_{m1})]}{(\omega^2 C_c C_L - G_{m1} G_{m2})^2 + \omega^2 C_c^2 (G_{m2} - G_{m1})^2}$

→  $v_{n,in}$



$C_c$   
 $v_{n,in}$   
 $v_o$   
 $C_L$

$G_{m1} (v_o - v_{n,in}) = (v_i - v_o) s C_c$   
 $(v_i - v_o) s C_c = G_{m2} v_i + v_o s C_L$   
 $\therefore v_{n,in} = \frac{v_o [s^2 C_c C_L + s C_c (G_{m2} - G_{m1}) + G_{m1} G_{m2}]}{G_{m1} (G_{m2} - s C_c)}$

$\therefore s v_{n,in} = \frac{s v_{n,o} [\omega^2 C_c^2 (G_{m2} - G_{m1})^2 + (G_{m1} G_{m2} - \omega^2 C_c C_L)^2]}{G_{m1}^2 (G_{m2}^2 + \omega^2 C_c^2)}$   
 $s v_{n,in} = \frac{s i_{n1}}{G_{m1}^2} \left[ \frac{G_{m2}^2 + \omega^2 C_c^2 (1 + G_{m2}/G_{m1})}{G_{m2}^2 + \omega^2 C_c^2} \right]$   
 $s v_{n,in} = s v_{n1} \left[ \frac{G_{m2}^2 + \omega^2 C_c^2 (1 + G_{m2}/G_{m1})}{G_{m2}^2 + \omega^2 C_c^2} \right]$



classmate

Date \_\_\_\_\_

Page \_\_\_\_\_

$s v_{in, in} = s v_{in} = \frac{4 \pi \gamma}{C_m} ; \omega \rightarrow 0$

2)

$\cdot g_{m1} v_T = (v_1 - v_T) s C_c$   
 $\cdot (v_1 - v_T) s C_c = g_{m2} v_1 + v_T s C_c$   
 $\therefore L(s) = \frac{v_T}{v_T} = \frac{g_{m1} (s C_c - g_{m2})}{s C_c (s C_c + g_{m2})}$

DC gain = 1  $\left\{ \begin{array}{l} \omega_{u, loop} = \frac{g_{m1}}{C_c} \\ z_1 = g_{m2} / C_c \end{array} \right\} \left\{ \begin{array}{l} p_1 = +g_{m2} / C_L \\ z_2 = g_{m2} / C_c \end{array} \right.$

3)

$$s v_{out} = \frac{\gamma \gamma}{C_L} \times \frac{4 g_{m1} C_L [g_{m2}^2 + \omega^2 C_c^2 (1 + g_{m2} / g_{m1})]}{\omega^4 C_c^2 C_L^2 + \omega^2 [C_c^2 (g_{m2} - g_{m1})^2 - 2 g_{m1} g_{m2} C_c C_L] + g_{m1}^2 g_{m2}^2}$$

$s v_{out} = s \omega_{in} / |H_1(s)|^2 + s \omega_{in} / |H_2(s)|^2$

$H_1(s) = \frac{s C_c - g_{m2}}{s^2 C_c C_L + s C_c (g_{m2} - g_{m1}) + g_{m1} g_{m2}} = \frac{(s C_c - g_{m2}) / C_c C_L}{(s + a)(s + b)}$

$H_2(s) = \frac{s C_c}{s^2 C_c C_L + s C_c (g_{m2} - g_{m1}) + g_{m1} g_{m2}} = \frac{s / C_L}{(s + a)(s + b)}$

where  $a + b = \frac{g_{m2} - g_{m1}}{C_L}$ ,  $a b = \frac{g_{m1} g_{m2}}{C_c C_L}$

classmate  
Date \_\_\_\_\_  
Page \_\_\_\_\_

$$\rightarrow H_1(s) H_1(-s) = \frac{(s c_c - \omega_{mv})(-s c_c - \omega_{mv}) \times \frac{1}{c_c^2 c_L^2}}{(s+a)(s+b)(s-a)(s-b)}$$

$$\cdot \text{residue 1} = \frac{(-a c_c - \omega_{mv})(-a c_c - \omega_{mv}) \times \frac{1}{c_c^2 c_L^2}}{(s=-a) \quad (-a+b)(-a-b) \times (-2a)}$$

$$\cdot \text{residue 2} = \frac{(-b c_c - \omega_{mv})(+b c_c - \omega_{mv}) \times \frac{1}{c_c^2 c_L^2}}{(s=-b) \quad (-b+a)(-b-a) \times (-2b)}$$

$$\cdot \oint |H_1(s)|^2 df = \oint H_1(s) H_1(-s) ds = \frac{\text{res}_1 + \text{res}_2}{2\pi j} = \frac{1}{4 c_c^2 c_L^2} \times \frac{1}{(a+b)} \left[ \frac{\omega_{mv}^2 + c_c^2}{ab} \right] = \frac{c_c \omega_{mv} + c_L \omega_{mv}}{4 c_c c_L \omega_{mv} (\omega_{mv} - \omega_{m1})}$$

$$\rightarrow H_2(s) H_2(-s) = \frac{-s^2 / c_L^2}{(s+a)(s+b)(s-a)(s-b)}$$

$$\text{res}_1 = \frac{-a^2 / c_L^2}{(a^2 - b^2)(-2a)} \quad \text{res}_2 = \frac{-b^2 / c_L^2}{(b^2 - a^2)(-2b)}$$

$$\therefore \oint |H_2(s)|^2 df = \frac{\text{res}_1 + \text{res}_2}{2} = \frac{1}{4 c_L^2} \times \frac{1}{a+b}$$

$$= \frac{1}{4 c_L (\omega_{mv} - \omega_{m1})}$$

$$\rightarrow \therefore r^2 = \sin \pi_1 \int_0^\infty |H_1(s)|^2 df + \sin \pi_2 \int_0^\infty |H_2(s)|^2 df$$

$$r^2 = \frac{\pi j [c_L \omega_{mv} + c_c (\omega_{m1} + \omega_{mv})]}{c_c c_L (\omega_{mv} - \omega_{m1})}$$

classmate

Date \_\_\_\_\_

Page \_\_\_\_\_

4)  $NBW = \frac{\omega^2}{s_{v_{n1}} \times 1^2} = \frac{g_{m1} [c_L c_{m2} + c_c (g_{m1} + g_{m2})]}{4 c_c c_L (g_{m2} - g_{m1})}$

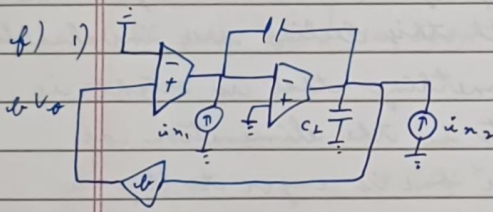
5)  $\omega_2 = \omega_1 = \omega_{u1} = \frac{g_{m1}}{c_c} \quad \left\{ \begin{array}{l} c_c = \alpha_c \times c_L \end{array} \right.$

$\therefore g_{m1} = c_c \omega_{u1} = \alpha_c c_L \omega_{u1}$

$\omega_2 = \omega_1 = \frac{g_{m2}}{c_L \omega_{u1}} \Rightarrow g_{m2} = \omega_2 c_L \omega_{u1}$

$\therefore s_{v_{n1}, in} = s_{v_{n1}} \left[ \frac{g_{m2}^2 + \omega^2 c_c^2 (1 + g_{m2}/g_{m1})}{g_{m2}^2 + \omega^2 c_c^2} \right]$

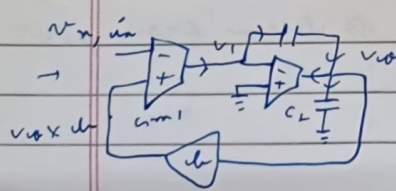
$s_{v_{n1}, in} = \frac{4kT}{\alpha_c^2 c_L \omega_{u1}} \left[ \frac{\omega_{u1}^2 \alpha_c^2 \alpha_c + \omega^2 \alpha_c^2 (\alpha_c + \alpha_2)}{\omega_{u1}^2 \alpha_c^2 + \omega^2 \alpha_c^2} \right]$

f) i) 

- $g_{m1} (v_{o1} + i_{n1}) = (v_{o1} - v_o) s c_c$
- $(v_{o1} - v_o) s c_c + i_{n2} = v_o s c_L + g_{m2} v_{o1}$

Solving,  $v_o = \frac{i_{n1} (s c_c - g_{m2}) + i_{n2} \times s c_c}{s^2 c_c c_L + s c_c (g_{m2} - g_{m1}) + g_{m1} g_{m2}}$

$\therefore s_{v_o} = \frac{s i_{n1} (\omega^2 c_c^2 + g_{m2}^2) + s i_{n2} \times \omega^2 c_c^2}{(\omega^2 c_c c_L - g_{m1} g_{m2})^2 + \omega^2 c_c^2 (g_{m2} - g_{m1})^2}$

ii) 

- $g_{m1} (v_{o1} - v_{n1}, in) = (v_{o1} - v_o) s c_c$
- $(v_{o1} - v_o) s c_c = g_{m2} v_{o1} + v_o s c_L$

classmate  
Date \_\_\_\_\_  
Page \_\_\_\_\_

Solving,  $V_o = \frac{G_m V_{in} (G_m - sC_c)}{s^2 C_c C_L + sC_c (G_m - G_m, dc) + G_m, G_m, dc}$

$\therefore V_{o, in} = \frac{V_{in} [(G_m, G_m, dc - \omega^2 C_c C_L)^2 + \omega^2 C_c^2 (G_m - G_m, dc)^2]}{G_m^2 (G_m^2 + \omega^2 C_c^2)}$

$\therefore V_{o, in} = \frac{V_{in} [G_m^2 + \omega^2 C_c^2 (1 + G_m/G_m)]}{G_m^2 (G_m^2 + \omega^2 C_c^2)}$

$= V_{in} \left[ \frac{G_m^2 + \omega^2 C_c^2 (1 + G_m/G_m)}{G_m^2 + \omega^2 C_c^2} \right]$

$V_{o, in} = V_{in} = 4 \times 7 / G_m, \quad ; \quad \omega \rightarrow 0$

2) ~~2)~~  $L(s) = \frac{dA(s)}{dc} = \frac{dG_m}{sC_c} \left( \frac{sC_c/G_m - 1}{sC_c/G_m + 1} \right)$

OC gain  $= \frac{1}{dc} \Big|_{\omega \rightarrow 0, \text{loop}} = \frac{dG_m}{C_c} \Big|_{\substack{f_z = G_m/C_L \\ z_c = G_m/C_c}}$

3)  $V_{out} = V_{in, 1} |H_1(s)|^2 + V_{in, 2} |H_2(s)|^2$

$H_1(s) = \frac{(sC_c - G_m)/C_c C_L}{s^2 + s(G_m - G_m, dc) + \frac{G_m, G_m, dc}{C_c C_L}}$

$H_2(s) = \frac{s/C_L}{s^2 + s(G_m - G_m, dc) + \frac{G_m, G_m, dc}{C_c C_L}}$



Using contour integration (same as part e)

$$\int |H_2(s)|^2 df = \frac{C_c G_{m1} b + C_L G_{m2}}{4 C_c C_L (G_{m2} - G_{m1} b) (G_{m1} b)}$$

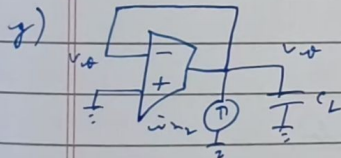
$$\int |H_2(s)|^2 df = \frac{1}{4 C_L (G_{m2} - G_{m1} b)}$$

$$\therefore \sigma^2 = \frac{K T [G_{m2} (C_L + C_c b (G_{m1} + G_{m2}))]}{C_L (G_{m2} - G_{m1} b) \times C_c b}$$

$$4) \text{ NBW} = \frac{\sigma^2}{S_{V_{in}} \times \left(\frac{1}{b}\right)^2} = \frac{b G_{m1} [C_L G_{m2} + C_c (G_{m1} + G_{m2})]}{4 C_c C_L (G_{m2} - G_{m1} b)}$$

$$5) C_c = a_c C_L \quad \left\{ \begin{array}{l} G_{m2} = a_2 C_L \omega_u \\ G_{m1} = A_0 \omega_u a_c C_L \end{array} \right.$$

$$\therefore S_{V_{in}, \omega} = \frac{4 K T}{A_0 \omega_u a_c C_L} \left[ \frac{1 + \omega^2 a_c a_2 / A_0}{a_2^2 \omega_u^2 + \omega^2 a_c^2} \right]$$



$$i_{nv} = G_{m2} v_{id} + v_{id} s C_L$$

$$\therefore S_{V_{in}, \omega} = \frac{S i_{nv}}{G_{m2}^2 + (2\pi f C_L)^2}$$

$$\therefore \sigma^2 = \int_0^\infty S_{V_{in}, \omega} df = \frac{K T}{C_L} \quad (\because S i_{nv} = 4 K T G_{m2})$$

classmate  
Date \_\_\_\_\_  
Page \_\_\_\_\_

· contribution of  $G_{m2}$  in d) =  $\frac{\chi J G_{m2}}{C_L (G_{m2} - G_{m1})}$

contribution of  $G_{m2}$  in f) =  $\frac{\chi J G_{m2}}{C_L (G_{m2} - G_{m1, b})}$

· If  $G_{m2} \gg G_{m1}, G_{m1, b}$ , contribution =  $\frac{\chi J}{C_L}$

from  $G_{m2} \rightarrow$  same as g)

6.2:

classmate  
Date \_\_\_\_\_  
Page \_\_\_\_\_

$$S_{I_{tail}} = \frac{2}{3} \frac{\gamma_T}{\gamma_T} \left[ g_{m1} + g_{m2} + g_{m3} + g_{m4} \right]$$

$$= \frac{16}{3} \frac{\gamma_T}{\gamma_T} \left[ g_{m1} + g_{m3} \right]$$

↓

Assuming  $g_{m1} = g_{m2}$   
and  $g_{m3} = g_{m4}$

1)  $M_1$ :-

- ↳  $i_{n1}$  flows down  $M_3$ , gets mirrored down  $M_4$ .
- ↳  $\frac{i_{n1}}{2}$  flows into  $M_3$ , gets mirrored into  $M_4$ , (and up  $I_{tail}$ )
- $i_{n1}/2$  flows up  $M_2$  and into  $I_{tail}$

$\therefore I_{n1} = i_{n1} \cdot \frac{i_{n1}}{2} + \frac{i_{n1}}{2} = i_{n1}$

2)  $M_2$ :-

- ↳  $i_{n2} = -I_{tail}$  directly
- ↳  $i_{n2}'/2 = I_{tail}$  through  $M_2$
- $i_{n2}'/2 = -I_{tail}$  up  $M_1$  and  $M_3 \Rightarrow$  up  $M_4$

$\therefore I_{n2} = -i_{n2}$

3)  $M_3$ :-

- ↳  $i_{n3}$  - no contribution to  $I_{tail}$
- ↳  $i_{n3}' = -I_{tail}$  (up  $M_3$  and mirrored up  $M_4$ )

CLASSMATE  
Date \_\_\_\_\_  
Page \_\_\_\_\_

$\therefore I_{n+3} = -i'_{n3}$

4)  $M_4$  :-  
 $\hookrightarrow i_{n4} = \text{no contribution}$   
 $\hookrightarrow i'_{n4} = I_{n+4} \text{ directly}$   
 $\therefore I_{n+4} = i'_{n4}$

$\therefore I_{n+} = i_{n1} + i'_{n4} - i_{n2} - i'_{n3}$   
 where  $S i_{n1} = S' i'_{n1} = \frac{8 \times 7}{3} \text{ gm}$   
 $\therefore S I_{n+} = \frac{16}{3} \times 7 (gm_1 + gm_3)$

(b)

Similar to part (a),



classmate

Date \_\_\_\_\_

Page \_\_\_\_\_

- 1)  $I_{n\phi_1} = i_{n1} + i'_{n1}/2 + i'_{n1}/2 = i_{n1}$
- 2)  $I_{n\phi_2} = -i_{n2} + i'_{n2}/2 - i'_{n2}/2 = -i_{n2}$
- 3)  $I_{n\phi_3} = -i_{n3} + 0 = -i_{n3}$
- 4)  $I_{n\phi_4} = -i_{n4} + 0 = -i_{n4}$

5), 6), 7), 8) currents of  $M_5$ ,  $M_6$ ,  $M_7$  and  $M_8$  circulate within those MOSFETs themselves. Essentially,  $i_{ni}$  and  $i'_{ni}$  cancel each other out, outside the MOSFET.

- 9)  $I_{n\phi_9} = 0 - i'_{n9} = -i'_{n9}$
- 10)  $I_{n\phi_{10}} = 0 + i'_{n10} = i'_{n10}$

$$\therefore S I_{n\phi} = \frac{8 \times 7}{3} [g_{m1} + g_{m2} + g_{m3} + g_{m4} + g_{m9} + g_{m10}]$$

$$= \frac{16 \times 7}{3} [g_{m1} + g_{m3} + g_{m9}]$$